

Write the Python program for Tic Tac Toe Game

Aim

To develop a Tic Tac Toe game in Python.

Objective

To implement a two-player Tic Tac Toe game and determine the winner.

Theory

Tic Tac Toe is a simple two-player strategy game played on a 3×3 grid. One player uses 'X' and the other uses 'O'. The players take turns marking empty cells. The objective is to place three identical symbols in a horizontal, vertical, or diagonal line before the opponent. If all cells are filled without a winner, the game ends in a draw. The program demonstrates concepts such as loops, conditional statements, lists, and game logic in Python.

Algorithm

1. Create a 3×3 board.
2. Display the board.
3. Allow players X and O to enter positions.
4. Update the board.
5. Check for winner after every move.
6. If board is full without winner, declare draw.
7. Display result.

Flowchart

Start

|

Create Board

|

Display Board

|

Player Move

|

Update Board

|

Check Winner

|

└──Winner?──┘

|

|

Yes

No

|

|

Display Continue

Winner Game

|

Stop

Python Code

```
board = [' ' for i in range(9)]
```

```
def display():
```

```
    print(board[0], "|", board[1], "|", board[2])
```

```
    print("--+---+--")
```

```
    print(board[3], "|", board[4], "|", board[5])
```

```
    print("--+---+--")
```

```
    print(board[6], "|", board[7], "|", board[8])
```

```
player = "X"
```

```
for turn in range(9):
```

```
    display()
```

```
    pos = int(input("Enter position (1-9): ")) - 1
```

```

if board[pos] == ' ':
    board[pos] = player

wins = [(0,1,2),(3,4,5),(6,7,8),
        (0,3,6),(1,4,7),(2,5,8),
        (0,4,8),(2,4,6)]

for w in wins:
    if board[w[0]] == board[w[1]] == board[w[2]] != ' ':
        display()
        print(player, "Wins!")
        exit()

player = "O" if player == "X" else "X"

```

```

display()
print("Game Draw")

```

Sample Output

```
X | O | X
```

```
--+---+--
```

```
O | X | O
```

```
--+---+--
```

```
X |  | 
```

X Wins!

Result

The Tic Tac Toe game was implemented successfully.

Conclusion

The program correctly allows two players to play and determines the winner or draw.

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Varshini M\AppData\Local\Programs
.py"
--+---+--
|  |
--+---+--
|  |
Enter position (1-9): 7
| X |
--+---+--
|  |
--+---+--
O |  |
Enter position (1-9): 1
X | X |
--+---+--
|  |
--+---+--
O |  |
Enter position (1-9): 3
X | X | O
--+---+--
|  |
--+---+--
O |  |
Enter position (1-9): 5
X | X | O
--+---+--
| X |
--+---+--
O |  |
Enter position (1-9): 6
X | X | O
--+---+--
| X | O
--+---+--
O |  |
Enter position (1-9): 8
X | X | O
--+---+--
| X | O
--+---+--
O | X |
X Wins!
```

```
> Users > Varshini M > OneDrive > Desktop > Here!!! > ARTIFICIAL INTELLIGENCE > VSC EXPs > EXP1.py
1  board = [' ' for i in range(9)]
2
3  def display():
4      print(board[0], "|", board[1], "|", board[2])
5      print("---+---+---")
6      print(board[3], "|", board[4], "|", board[5])
7      print("---+---+---")
8      print(board[6], "|", board[7], "|", board[8])
9
10 player = "X"
11
12 for turn in range(9):
13     display()
14     pos = int(input("Enter position (1-9): ")) - 1
15
16     if 0 <= pos <= 8 and board[pos] == ' ':
17         board[pos] = player
18
19         wins = [(0,1,2), (3,4,5), (6,7,8),
20                 (0,3,6), (1,4,7), (2,5,8),
21                 (0,4,8), (2,4,6)]
22
23         for w in wins:
24             if board[w[0]] == board[w[1]] == board[w[2]] != ' ':
25                 display()
26                 print(player, "Wins!")
27                 exit()
28
29         player = "O" if player == "X" else "X"
30
31     else:
32         print("Invalid position or already occupied. Try again.")
33         continue
34
35 display()
36 print("Game Draw")
```